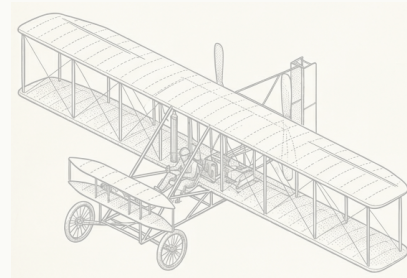


1903

FIG. 01



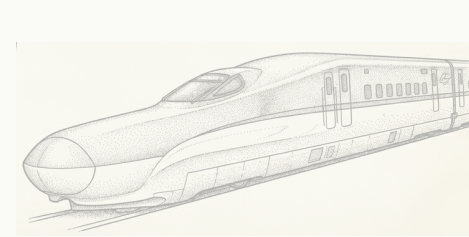
BICYCLE



FLYING MACHINE

1997

FIG. 02



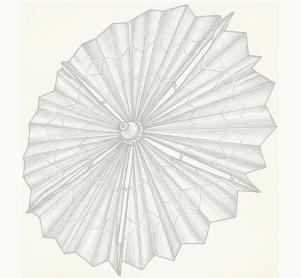
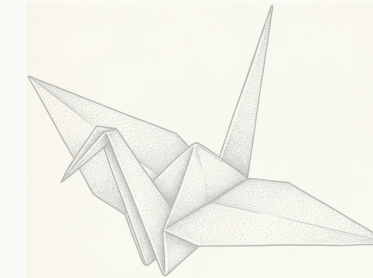
KINGFISHER BEAK



SHINKANSEN NOSE

2016

FIG. 03



ORIGAMI



SOLAR SAIL

Carnegie  
Mellon  
University



Human-  
Computer  
Interaction  
Institute

KITTUR LAB · HUMAN-COMPUTER INTERACTION INSTITUTE

# Solving industry's hardest problems with solutions from *unexpected places*.

What the research offers, which companies to bring us, and how it can help you open doors.

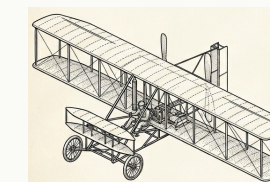
THE PROBLEM EVERY COMPANY BRINGS

Every R&D team is stuck on problems they're making only *incremental progress* on.

Because experts search where they already know — and AI gets you more of what's already out there.

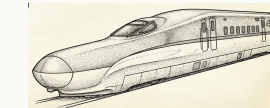
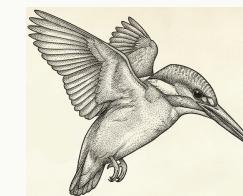
Our research makes finding breakthroughs from distant domains *systematic*: retrieve by mechanism, not keyword; map it into their domain; de-risk it into something their team can build and test.

BREAKTHROUGHS COME FROM ANALOGIES



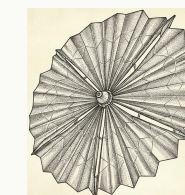
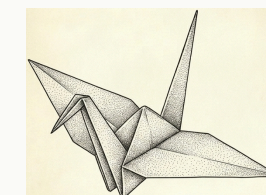
Bicycle → Wright Flyer (1903)

Frame-lean steering became wing-warp steering: the control problem that grounded every prior aviator.



Kingfisher → bullet train (1997)

A beak that enters water without a splash became a nose that ends sonic booms in tunnels.



Origami → NASA solar sail (2016)

A Miura fold packs a 25 m sail into a 1 m fairing: a packaging problem became a paper-folding one.

Serendipity found these. We make them searchable — forward, and in reverse (a capability → the problems it could solve).

12+ years of research at HCII    PNAS · best papers at CHI, CSCW, KDD

NSF-funded · 4 patents    Kittur · Martelaro · Mohanty · Shahaf

WHAT IT LOOKS LIKE · 60 SECONDS

# A public P&G challenge: *clean between teeth without floss.*

Nobody in oral care would look at an oil rig. The engine did – and came back with a concept, its evidence, and the tests to run.



## 01 SEARCH

Retrieve by *mechanism*: “dislodge a stubborn deposit from a narrow channel.” Matches came from **drilling fluids** that scour a borehole and **vascular medicine**, where magnetic bead swarms clear clots.

## 02 TRANSFER

Map the mechanism onto the real constraints: safe in the mouth, no hardware a consumer won’t use, a sachet and a rinse. Every claim is tied to a primary source – not a hallucinated one.

## 03 EVALUATE

Hand back the four test protocols and success criteria that settle the risk: chain formation, biofilm softening, debris removal, bead recovery. The team knows what to run Monday.

The same loop runs on a materials scientist’s holy-grail problem or a propulsion chemist’s storage problem – with their experts in the loop.

PROOF · WITH INDUSTRY PARTNERS

# Ideas their own experts rated as ones *they couldn't have reached.*

COVESTRO · FORTUNE 500 MATERIALS

3 “HOLY-GRAIL” PROBLEMS STUCK 10+ YEARS · 17 SCIENTISTS

54%

more novel, no drop in feasibility (p < .05)

30+

promising ideas vs. 0–1 from their own LLM tools

2

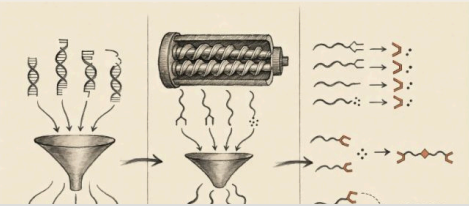
already in their managed innovation pipeline

IDEA 01 OF 03NEXT >

BORROWED FROM DNA-SEQUENCING PREPARATION

Convergent chain repair: boron funnels every broken end into one joinable handle

Borrow the terminus-funneling step from next-generation sequencing library preparation to converge every broken polyolefin chain end onto one boron handle and rebuild linear molecular weight without radicals or gel.



HOW IT WORKS

Thexylborane, carrying exactly two B–H bonds, hydroborates terminal vinyl and vinylidene chain ends in the melt, capturing two chains per molecule onto one dialkylborane handle; a solid-acid co-additive funnels carbonyl-derived ends into the same pathway via dehydration. Downstream CO injection drives Brown carbonylation ③, migrating both chains onto a carbonyl carbon to yield a linear ketone bridge – purely ionic, no radical generated, mid-chain attack impossible. No one has combined

“If we sat down and spent a week going through these things, that would be a worthwhile investment to make sure that we’re spending the next two years on the right projects.” — R&D group leader

URSA MAJOR · AEROSPACE PROPULSION “HOW CAN WE EXTEND THE LIFE OF OUR PROPELLANT?”

+80%

estimated propellant life at minimal extra cost vs. 5–10% from their previous approach

Paid pilot underway.

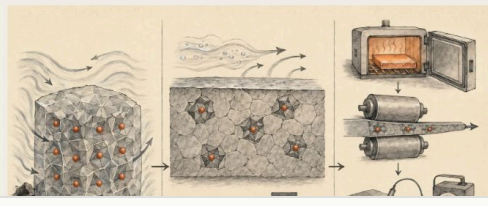
Concepts borrowed from semiconductor wafer gettering and nuclear-waste ceramics; scope expanding.

IDEA 03 OF 03< PREV

BORROWED FROM NUCLEAR-WASTE CERAMICS

Lattice-pinned alloy: heat treatment locks trace metals where peroxide cannot reach

Borrow lattice-site immobilization from nuclear-waste ceramics to trap catalytic trace metals in aluminum tank plate into crystal sites hydrogen peroxide physically cannot breach, enabling decade-scale storage without hazardous passivation.



HOW IT WORKS

Annealing 5254 plate at 480 °C drives residual Fe and Cu into Al<sub>3</sub>Fe and Al<sub>6</sub>(Fe,Mn) intermetallics ②⑤, ③② whose formation enthalpies exceed the oxidation potential H<sub>2</sub>O<sub>2</sub> can deliver, locking catalytic metals in lattice sites the oxidizer cannot breach. Current practice controls only composition and surface chemistry ③④; this opens a third axis – microstructural site occupancy – set by a single upstream heat treatment before cold-rolling to flight strength. Intermetallics formed above 400 °C are

“If this was [available to every engineer] they would [use it] daily.” “...ideas which no one in my team could come up with.” — R&D leader, Fortune 500 automotive

BUILT ALONGSIDE Covestro · Ursa Major · Toyota · Bosch · Conservation X Labs · and a public P&G challenge

CONFIDENTIAL

WHO TO BRING US

Industrial R&D with a *named, stuck problem* – or an underused capability.

partner in conversation plain: target

MATERIALS & CHEMICALS

Covestro

BASF

PPG

Dow

DuPont

3M

Henkel

Solvay

Evonik

SABIC

Arkema

AUTOMOTIVE & MOBILITY

Toyota

Bosch

GM

Ford

Stellantis

Honda

Magna

Cummins

CONSUMER GOODS & FOOD

P&G

PepsiCo

Chobani

Unilever

Nestlé

Mondelez

Mars

L’Oréal

Kraft Heinz

Colgate-Palmolive

ENERGY & INDUSTRIAL

Westinghouse

Wabtec

Alcoa

GE Vernova

Siemens Energy

SLB

Eaton

Caterpillar

AEROSPACE & DEFENSE

Ursa Major

RTX

Lockheed Martin

Northrop Grumman

Boeing

Honeywell

Anduril

Blue Origin

ELECTRONICS & SEMICONDUCTORS

Mitsubishi Electric

Intel

Micron

Corning

Applied Materials

Samsung

TSMC

## HOW TO SPOT A FIT

# The right person, the signal, and *the question that opens it.*

## 01 THE RIGHT PERSON

VP or Head of R&D

CTO · Chief Science or Chief Innovation Officer

Open-innovation and alliance-management leads

The BD lead looking for new markets for an existing material or IP

Not the CIO, and not procurement — the person who owns the technical roadmap and the problems on it.

## 02 THE SIGNAL IT'S A FIT

A problem their team has been on for years

A paid-for capability whose best markets they can't see

They've tried the LLMs and been underwhelmed

An open-innovation mandate and appetite for a pilot

One of these is enough. Two and it's a strong lead.

## 03 THE QUESTION THAT OPENS IT

*“What's the problem your best people have been on the longest?”*

*“What have you paid to develop that you suspect has uses you can't see from here?”*

*“Where did the AI tools your teams tried fall short?”*

Send us the answer. We come back with directions and the CMU people who could pursue them.

## HOW THIS CAN HELP YOU

# Walk into the first meeting with **three concrete directions** — and the CMU people who could pursue them.

## 01 Start the conversation

A company's interest is nebulous, or they name a problem they've been stuck on. Send it to us before the first meeting. You walk in with three concrete directions on *their* problem — a reason to meet, not a brochure.

For production deployment inside a company there is also a CMU spinout, Analogical Engines, built on this research — so an inquiry can be routed to the right door: exploratory work through the lab, deployment through the company.

## 02 Sponsored research & workshops

Through the lab: a half-day ideation session with their R&D team on a named problem, or a scoped sponsored project on the thing they've been stuck on. Their experts stay in the loop; our students and postdocs do the work.

## 03 Plan the outreach

Before you pick up the phone: from a company's capabilities and IP, we map which labs and projects across campus they could fund or license — so you know which faculty to pair with which company, and who to call on next.

## BRING ME A COMPANY AND A PROBLEM

[nkittur.github.io/unexpected-places](https://nkittur.github.io/unexpected-places)

Case studies, what we offer, who to bring · [nkittur@cs.cmu.edu](mailto:nkittur@cs.cmu.edu)

Happy to talk to any of you.

Niki Kittur